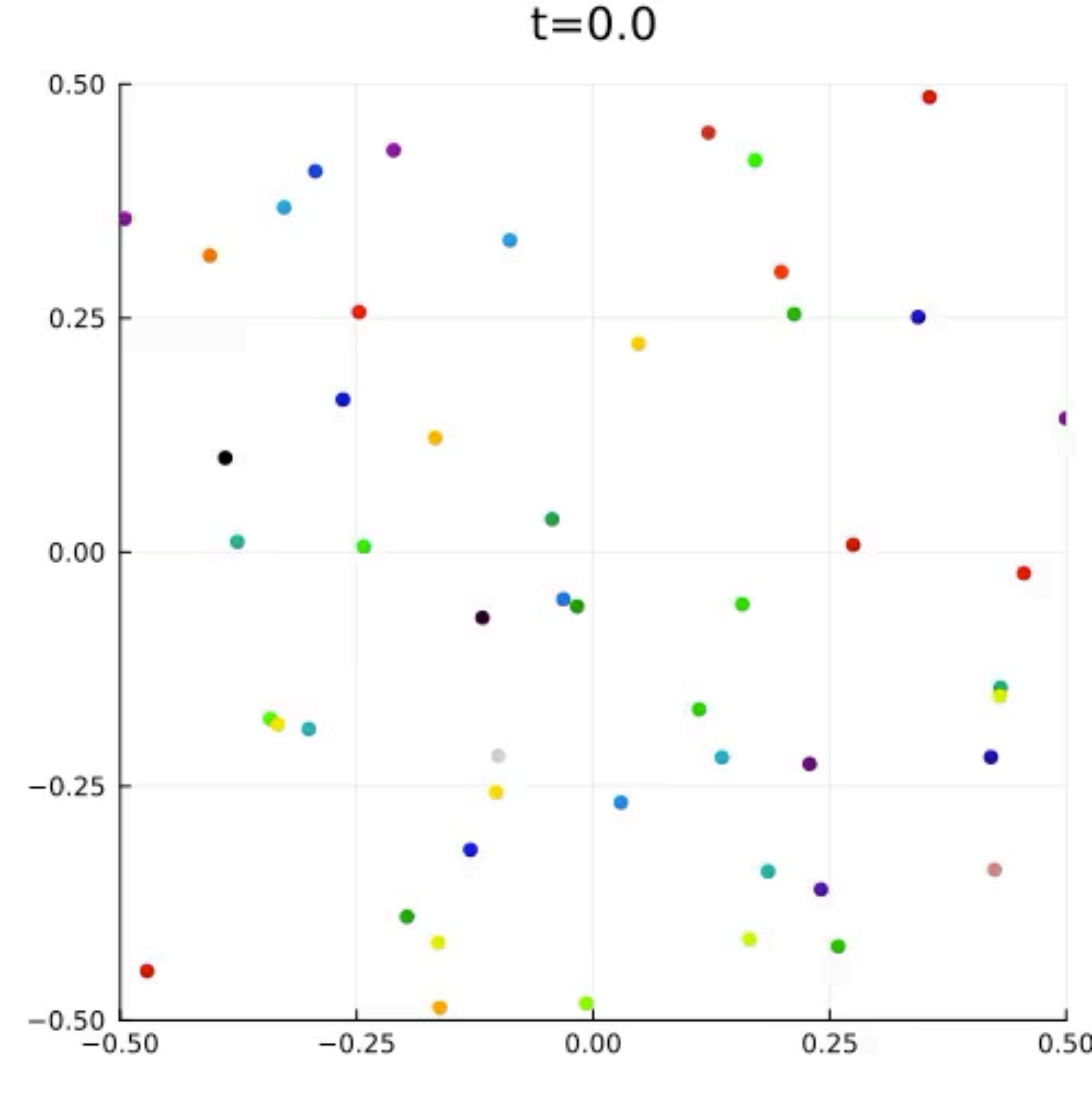
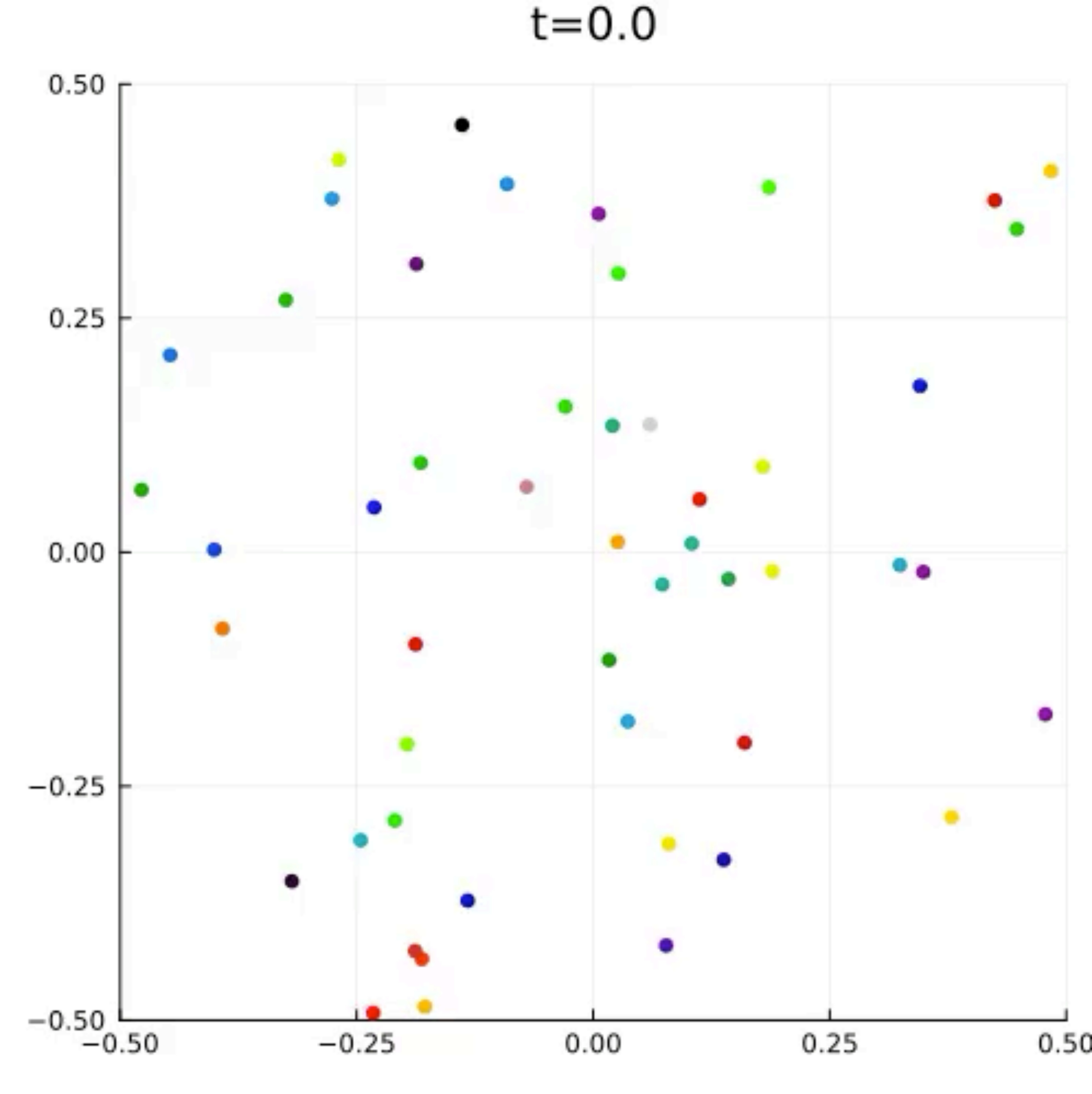
# Active ants

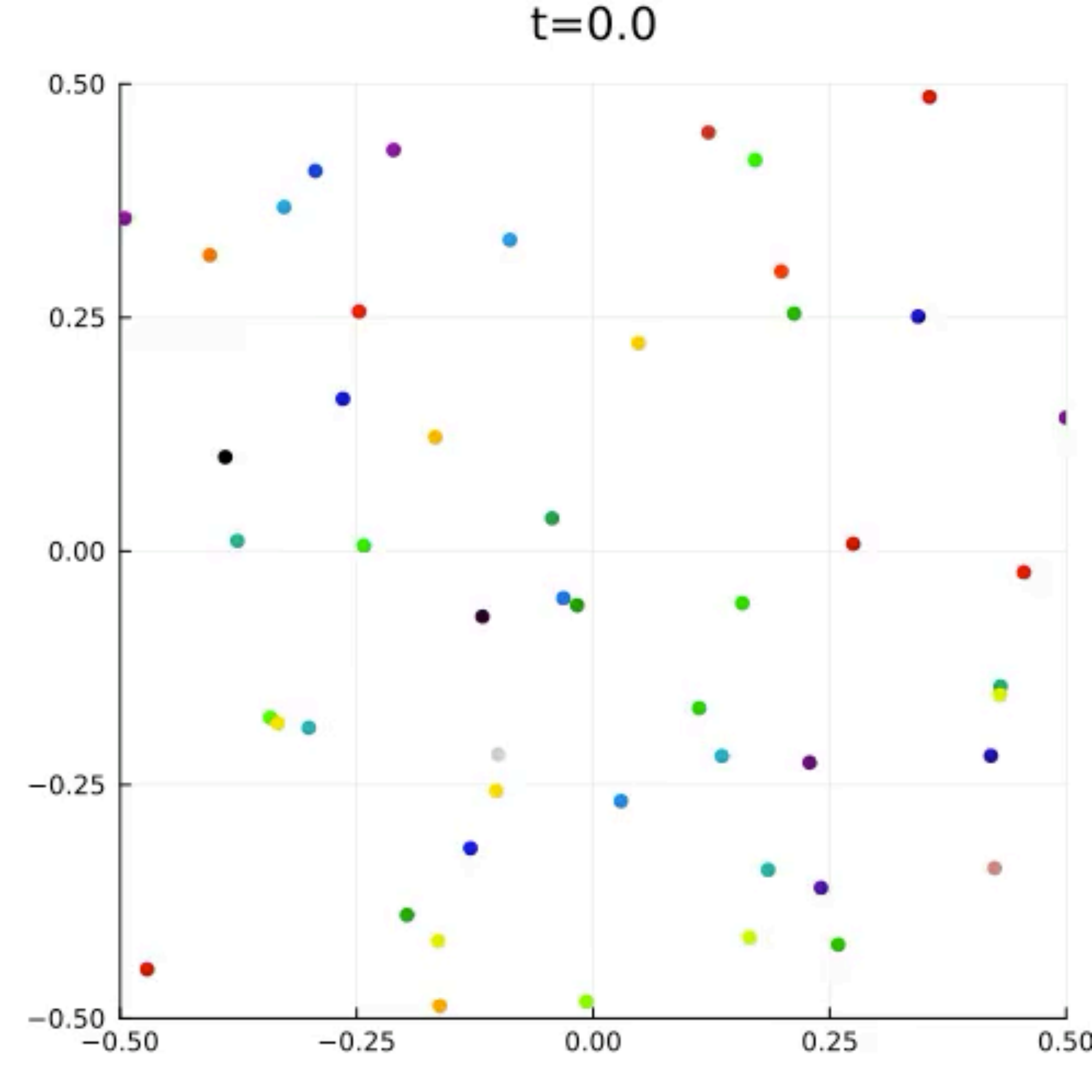
- $F = n(\theta_t^i) \cdot \frac{1}{N} \sum_{j \neq i} \nabla K(X_t^i + \tau v(\theta_t^i) - X_t^j)$ on $\mathbb{T}^2$

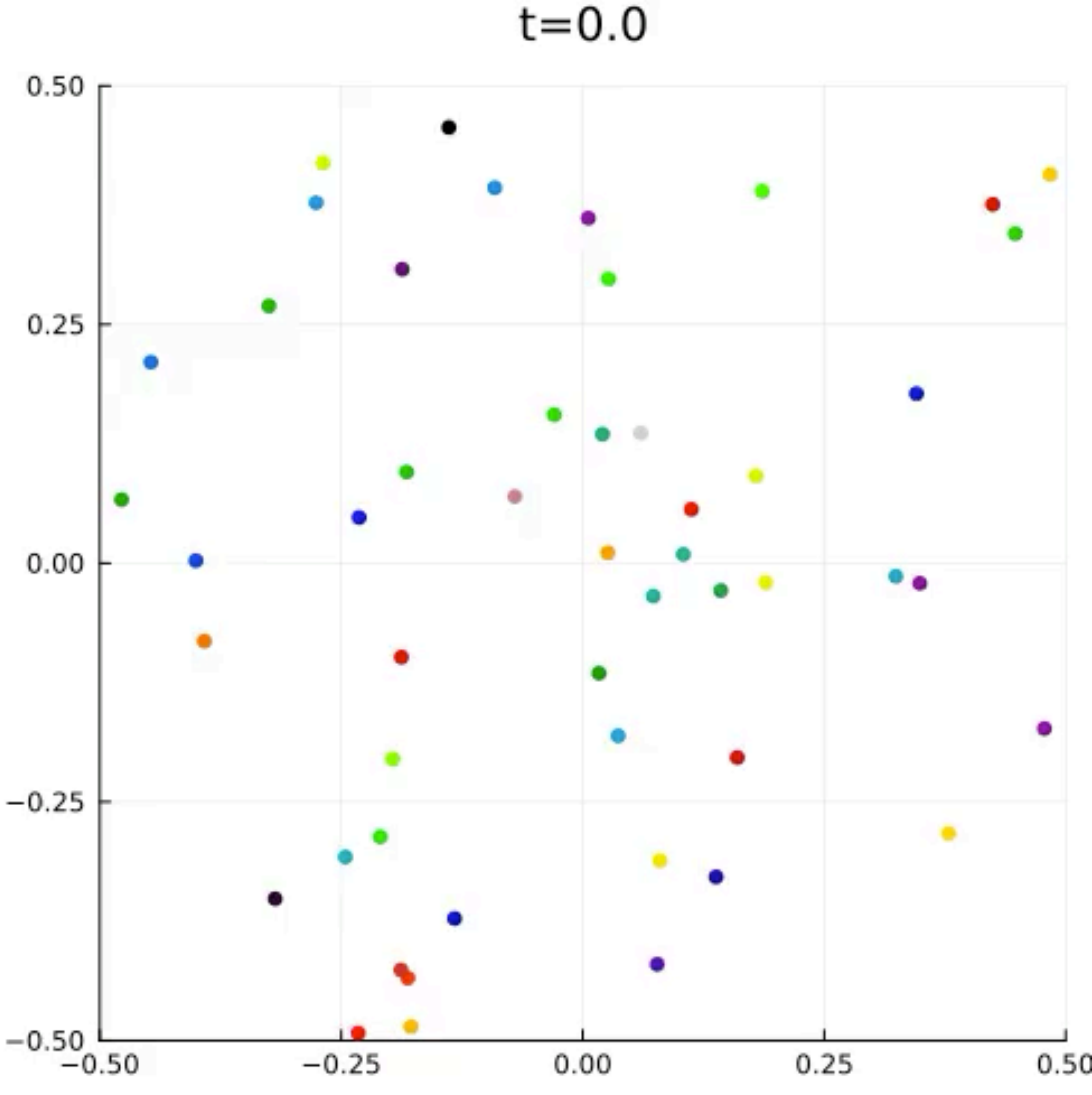

t=0.0
0.50
0.25
0.00
−0.25
−0.50
−0.50
−0.25
0.00
0.25
0.50

$$\tau = 0$$


t=0.0
0.50
0.25
0.00
−0.25
−0.50
−0.50
−0.25
0.00
0.25
0.50

$$\tau = 0.15$$


t=0.0
0.50
0.25
0.00
−0.25
−0.50
−0.50
−0.25
0.00
0.25
0.50


t=0.0
0.50
0.25
0.00
−0.25
−0.50
−0.50
−0.25
0.00
0.25
0.50

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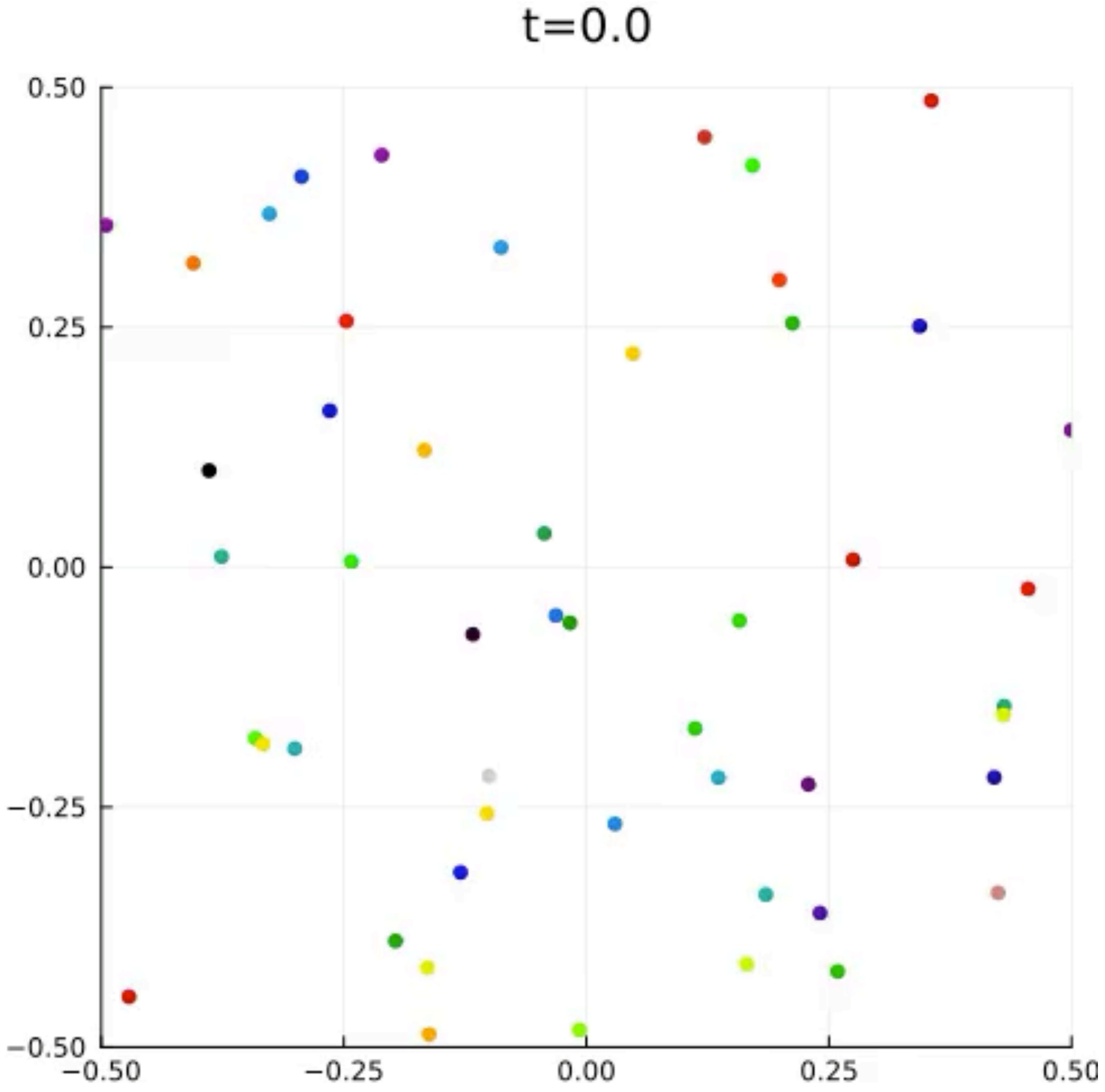
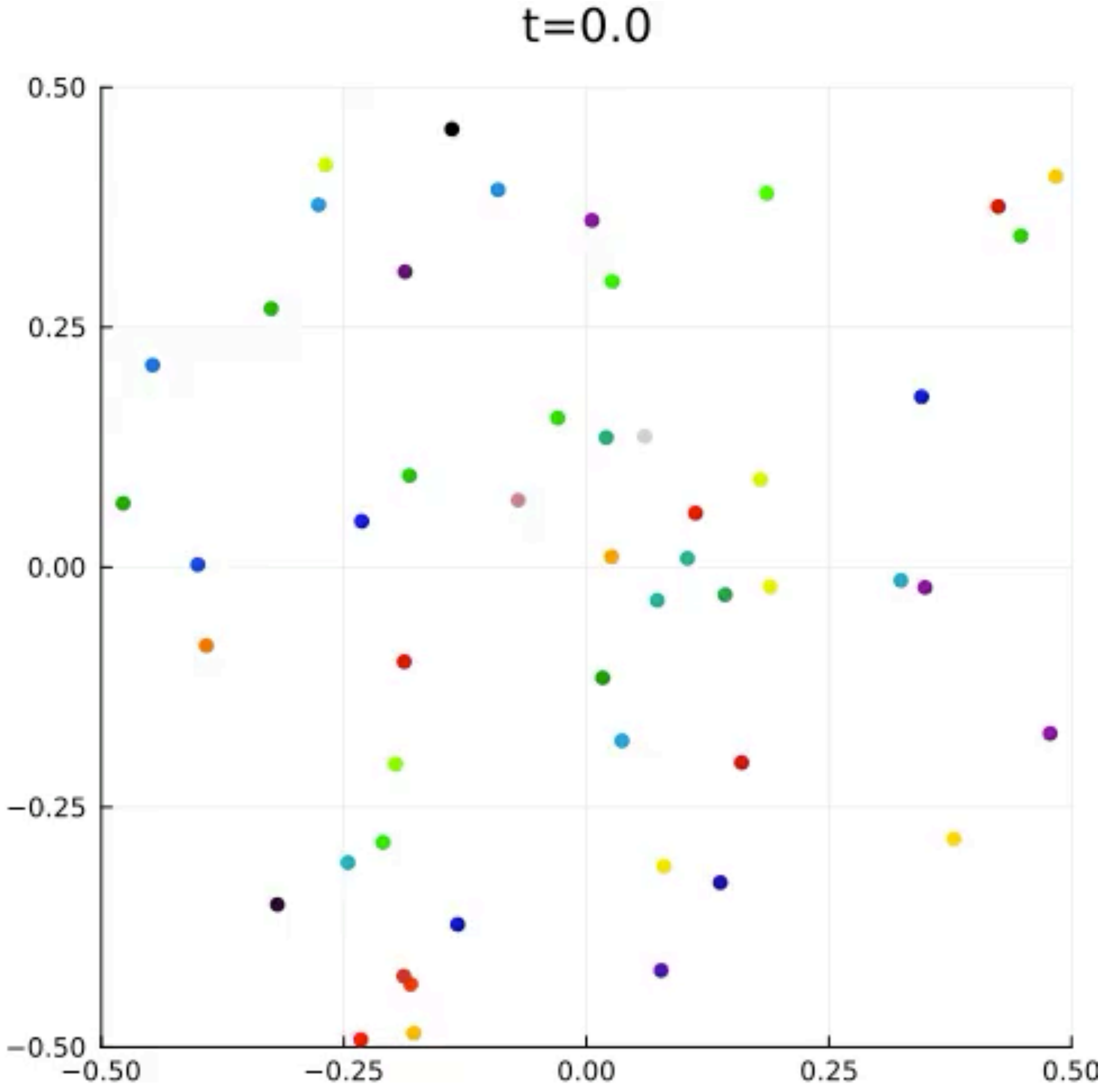



$\tau = 0$




$\tau = 0.15$

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- Anticipation length $\tau$ as a phase transition/bifurcation parameter?